

LC-MARE

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Objective

Aim of the low-cost science-grade marine autonomous acoustic recorder (LC-MARE) is to provide a cost-effective solution to access and record underwater sound. The implementation should be low-cost but without limitations on sound quality.

The project is completely open source, and all information may be found on [GitHub](#).

Concept

The implementation should be low-cost but without constraints on scientific data quality.

An underwater autonomous recorder consists of four major components:

- Hydrophone
- Recording electronics
- Batteries
- Pressure housing

While the electronic components are nowadays easily available and relatively cost-effective, hydrophones and pressure housing are notoriously expensive but important components, as they determine the scientific quality of the collected data. Therefore, it is essential to define upfront the use-case and decide on:

- Minimal useful acoustic bandwidth needed for the expected scientific results as this defines the size of the hydrophone.
- Maximal deployment depth must be considered for the hydrophone selection and pressure housing material and construction.
- Endurance of the recorder, as this in connection with the power consumption of the electronics determines the size of the required battery pack.

Acoustic bandwidth :

The acoustic bandwidth of the recorder is a significant cost factor. It seems clear that use cases that are only concerned with signals, say below 5 kHz, the use of hydrophones and recorder bandwidth of 100 kHz or more do not add to the relevance of the acoustic data. Using an acoustic bandwidth of 20 to 40 kHz and a sampling frequency of 48 kHz are more than adequate. However, use cases where dolphin or beaked whale echolocation clicks are the signals of interest, broadband hydrophones (i.e. bandwidth >100kHz) and sampling frequencies of 192 kHz are asked for.

As sampling frequencies and acoustic bandwidth are related, so does the required storage increase with the sampling frequency. As a side-effect increased sampling frequency requires also extended memory to buffer the recorded data during the archiving process.

Deployment depth;

The deployment depth of the recorder can have a significant impact on its implementation, especially if the recorder contains removable components, like microSD cards, batteries or connectors, or movable components, like buttons. In these cases, the recorder housing must contain air and therefore withstand

a pressure difference that corresponds to the deployment depth. Only recorders without (re)movable components can be filled with oils or resins.

Not only electronic containers must withstand ambient pressure, but also hydrophones that to achieve best performance must contain air. From the two typical hydrophone forms, cylindrical and spherical, only the spherical form is designed for great deployment depth well beyond 1000 m, while cylindrical hydrophones may crash well below 1000 m. Only special hydrophone encapsulation may increase the hydrophone survival depth to well above 1000 m.

Recorder endurance:

As with all autonomous systems the sound recorder must contain its own power source, which in most cases is provided in form of one or more battery packs. To assess the endurance of the recorder the consumption must also be known. If continuous recording is desired, then the recorder consumption is the limiting factor, and the size of the power source is a consequence of the recording time. If duty-cycling of the recording is acceptable, then the recorder endurance is a consequence of battery capacity, power consumption and recording schematic.

Aim and use cases of LC-MARE

The envisioned use cases are defining the required characteristics of the LC-MARE autonomous recorder, and LC-MARE is aimed at generation of science-grade marine acoustic data that support

- entry-level scientific research (e.g. for undergraduate and graduate students)
- environmental monitoring of coastal marine areas (being protected or not)
- marine bioacoustics in STEM education

It is envisioned that LC-Mare

- is used up to a recorder depth of 100 m.
- is recording up to 50 kHz bandwidth.
- Is continuously recording acoustic data up to 30 days.
- may be calibrated to allow comparisons and exchanges of data.

Cost factors

In addition to the envisioned use cases, the convenience of the LC-MARE implementation is determined by its complexity and cost factors. If the implementation is too complex or too costly a custom implementation may not be convenient.

There are different levels of complexity and cost factors.

- Commercially available components:
 - Piezo ceramic tubes
 - Programmable recording system (MCU, microSD card, RTC)
 - Pressure housing components.

These components are typically produced in quantities, and no major cost saving is anticipated.

- Specific made components:
 - Analog amplifier
 - Analog-digital conversion
 - Base board (to avoid cabling)

These components are specific made for LC-MARE and the 'set-up' costs are usually higher than the component costs. Therefore, only the production of multiple units (say 50 to 100+units) reduces the price-tag per unit to nearly the component costs.

- Custom made components:
 - Hydrophone construction
 - Pressure vessel construction.
 - Mounting frame

Depending on the use cases and individual skills these components may be built by the interested users, whereby the hydrophone construction is the most demanding task and may be appropriate only for experienced users. It is expected that, while DIY is possible, for most users the hydrophones are constructed by specialized businesses.

The construction of pressure vessels depends again on the use-cases and assuming shallow water applications (recorder depth <100 m) user build pressure vessels are feasible using pressure-rated PVC tubes and fittings.

Implementation

Hardware LC-Mare V2

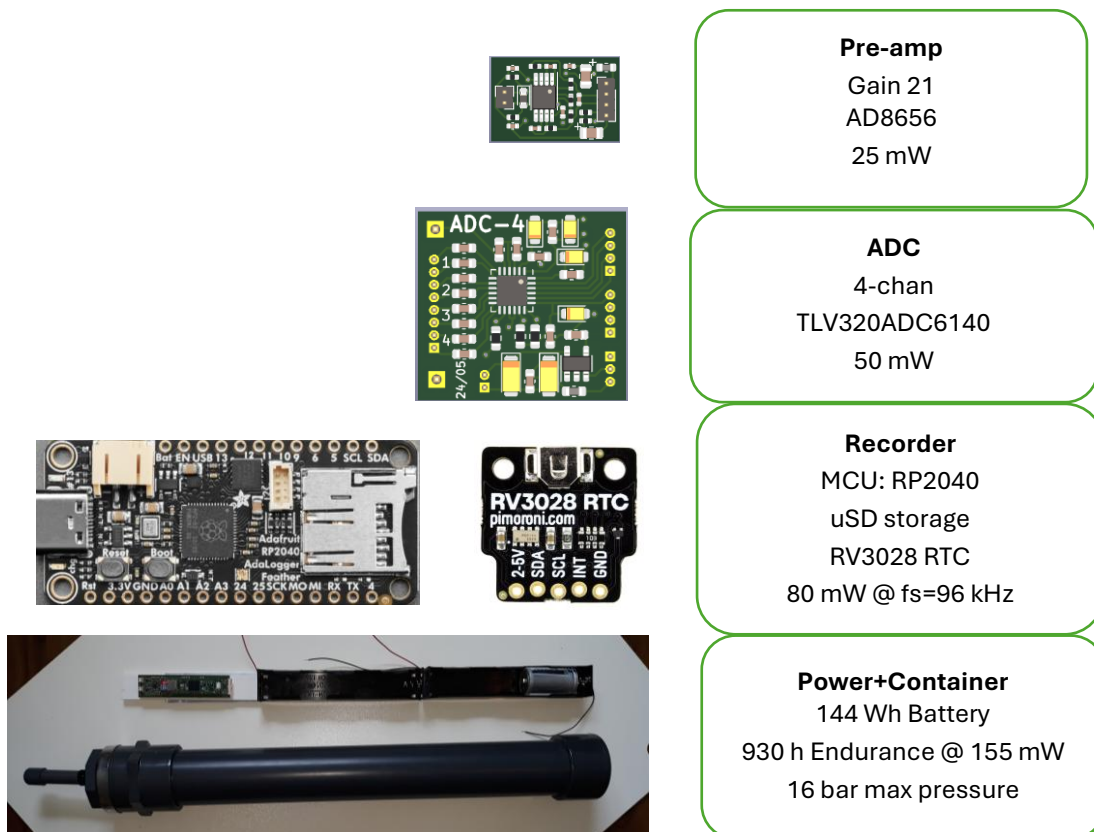


Figure 1: LC-Mare V2 Hardware components.

The components of a first LC-MARE prototype are shown in Figure 1. The electronics is composed of multiple components to facilitate easy replacement of components, if deemed necessary.

Length and diameter of the pressure tube are determined by the size of the battery holders and resulted to 500 x 63 mm for a 6 D-Cell battery pack. Using additional fittings and tubes, the power supply can easily be extended by adding additional four battery containers.

The pressure tube and fittings are 16 Bar rated PVC tubes, commercially made for water pipe constructions, e.g. for swimming pools.

Software

The software is using standard C/C++ code and by being compatible with the Arduino infrastructure it is easily maintained, modified and downloaded to the microcontroller of the LC-MARE. The few parameters that are expected to be modified at sea may be provided using a terminal program, a python GUI or by a file on the microSD card that can be prepared prior to the at-sea activity.

Recorder



Figure 2: LC-Mare August 2024 version.

Implementation details

The LC-MARE implementation will be open source and hard and software implementation will be published at [GitHub - WMXZ-EU/LC-MARE: low-cost marine acoustic recorder](#)

Hydrophone

Idea of a custom hydrophone



Figure 3: Custom made hydrophone using piezo-ceramic tubes

This Hydrophone consists of three components: Cylindrical piezo-ceramic tube, BlueRobotics penetrator, and polyurethane rubber. The hydrophone is at some distance from the penetrator to reduce acoustic interference of the pressure container. The use of BlueRobotics penetrators facilitates the changes of hydrophones and pressure containers.

As Endcapping of piezo-ceramic tubes is somewhat demanding it may be necessary to obtain ready-to-use COTS hydrophones, which are then mounted onto BlueRobotics penetrators as shown in Figure 4



Figure 4: LC-Mare hydrophone SQ-17 potted onto BlueRobotics penetrator.

Pressure Container

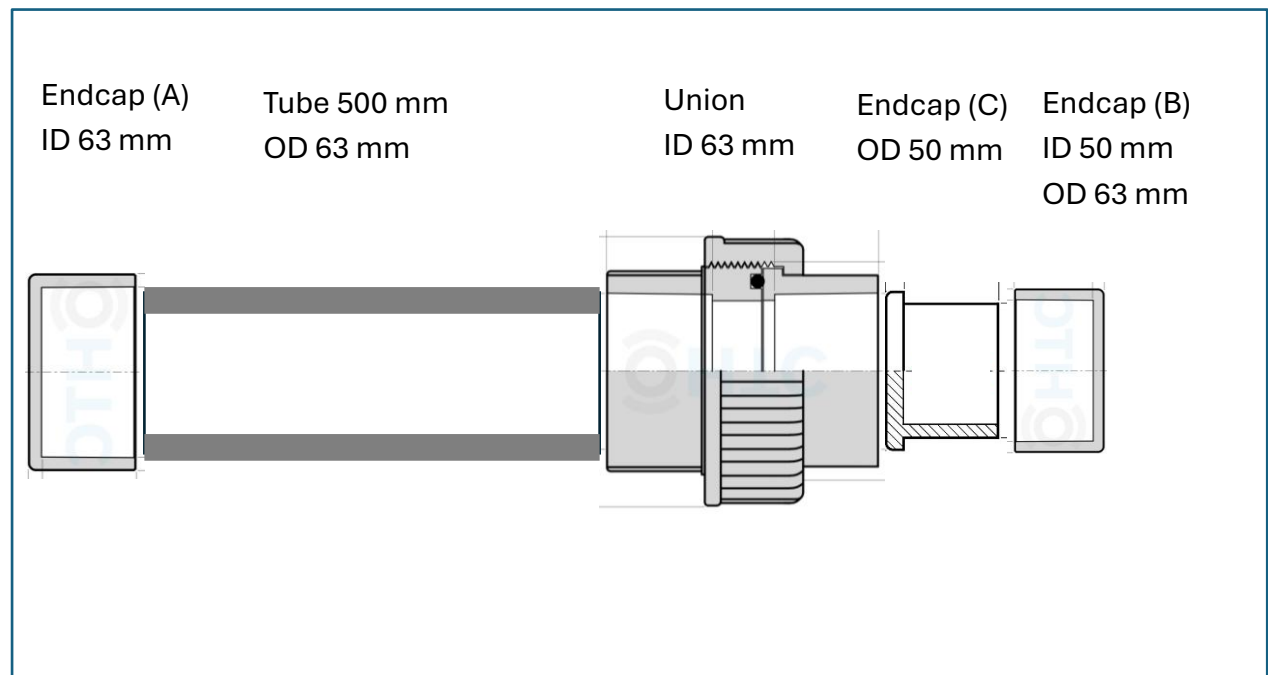


Figure 5: Pressure container components.

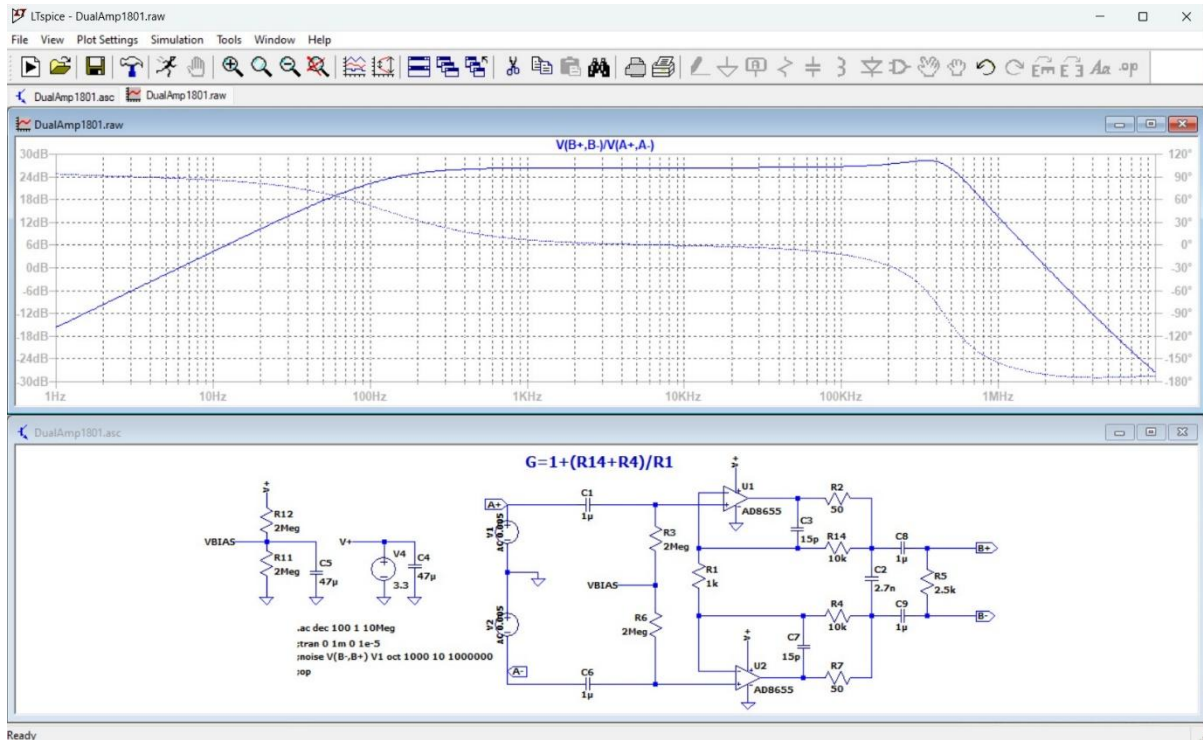
The pressure container consists of 5 components a 500 mm long tube, one union, 3 Endcaps with different diameters. To connect the hydrophone a 10 mm hole will be drilled in the middle-sized endcap (Endcap (B)). The PVC-U components are rated with PN16 and should withstand 16 bar internal pressure. It is expected that they also withstand 15 Bar external pressure, which corresponds to a water depth 150 m. The PVC-U tubes and fittings are purchased from an online store (pcv-welt.de) but may be found in hardware stores.

Electronics

For the electronics we have following components

Pre-amplifier

The pre-amplifier is a differential-to-differential voltage amplifier, the Spice simulation is shown next:



Gain is controlled effectively by the resistor R1. There is still some gain increase at 400 kHz, which can be removed by increasing the feedback capacitors C3, C7 somewhat. The selected OP-AMPS are of CMOS type with 2.7 nV/sqrt(Hz) noise voltage each.

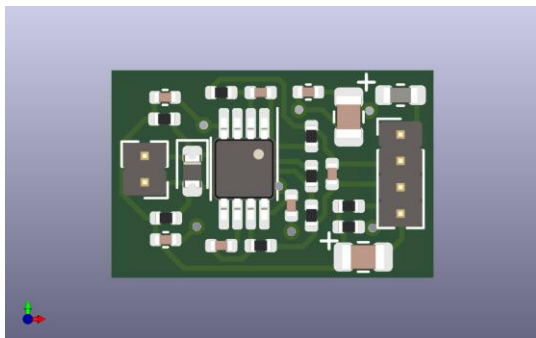


Figure 6: Preamplifier

A more lower noise pre-amplifier with FET and ultra-low noise bipolar amplifier would be a possible alternative and is in development.

Analog Digital Conversion

The ADC is based on the Texas Instruments TLV320ADC6140 chip and implements the circuitry of the datasheet.

The ADC features 4 differential input channels each of which can be amplified up to 40 dB. A dedicated LDO regulator provides 3.3V for the analog circuitry.

Sampling frequencies can be up to 768 kHz for a single channel, 384 kHz for two channels and 192 kHz for all four channels. Bit depth is 24 bits.

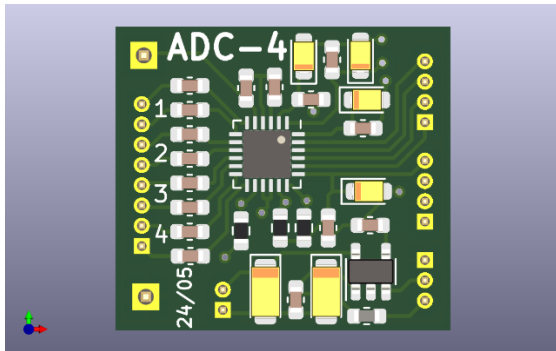


Figure 7: Analog-digital-converter

Putting it all together

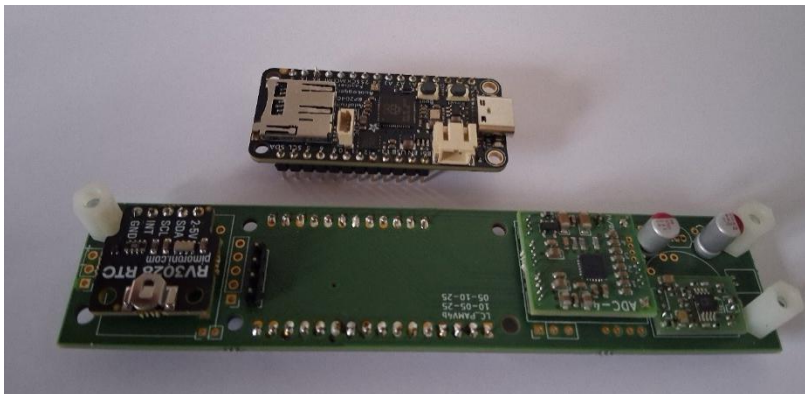


Figure 8: All components and connecting base board

The final electronics board features 1 pre-amp, a 4-channel ADC, an Adafruit Feather Adalogger Rp2040 and a RV3028 RTC.

Software

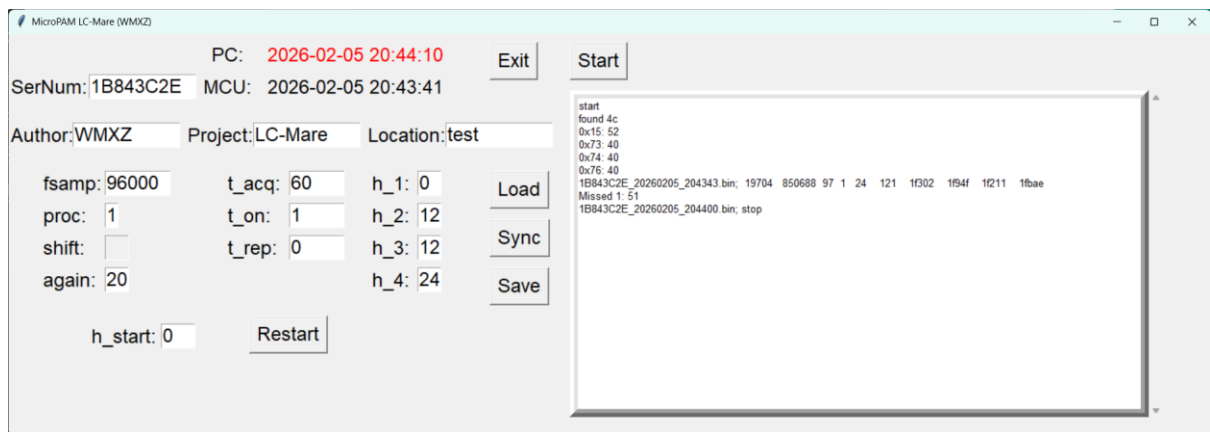
Fully programmed with C/C++ code but can be compiled and downloaded within the Arduino ecosystem or using Makefiles.

All software is available on [Github](#).

Data storage is using ExFat filesystem and after removing the disk, the data can be accessed directly by the PC. This allows also to store/edit configuration files on the disk ahead of closing the pressure container.

GUI

The following Python based GUI shows some of the parameters that may be retrieved and stored and may be the basis for a Custom GUI.



This GUI shows some parameters that are used for controlling the functionality of LC-MARE. Gray-ed fields are de-activated as not in use in the actual implementation.

The Form will be populated with the 'Load' button. The 'Sync' button will synchronize the RTC with the PC time. Parameters will be downloaded to MCU using the 'Save' button.

The start button will start the acquisition process and will display the messages from the MCU. At the same time the button changes to a Stop button to stop the acquisition process.

The Restart button will reboot the connected MCU. If the h_start value is zero, then the MCU will reboot immediately. Any value >0 will delay the reboot to the next full hour. For example, for a value of 1 the MCU will reboot at the next full hour, with a value of 5 the MCU will reboot at the last full hour in the next 5 hours.

Parameters shown in GUI:

- fsamp: sampling frequency in Hz
- proc: online processing: 0 wav file; 1: integer loss-less compression.
- shift: in case of compression indicates the number of lower bits that are to be ignored (deactivated in screenshot)
- again: analog gain in dB in ADC

- t_acq: length of files in s
- t_on: length of acquisition block in s
- t_rep: time in s between starts of acquisition blocks in s (0 means continuously)

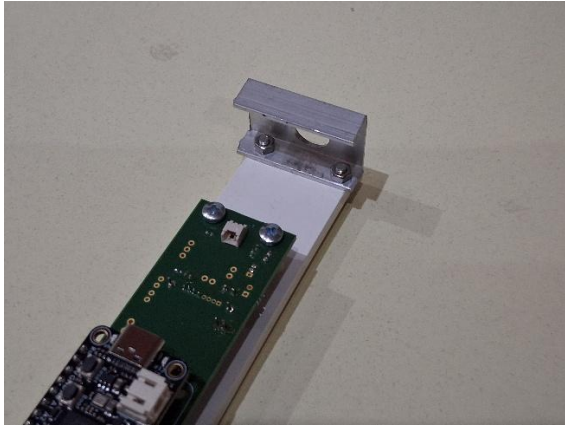
- h_1: start hour of first acquisition period
- h_2: end of first acquisition period
- h_3: start of second acquisition period
- h_4: end of second acquisition period

- h_start: reboot delay in hours (0 immediately, 1 next full hour, etc)

In above GUI image, the acquisition timing parameters are set for continuous recording.

Mounting system

The electronic and battery packs are mounted onto an aluminium plate (35x50x2) by means of double-sided adhesive tapes. To separate electronics from conducting aluminium an additional PVC plate is added. The mounting plate is attached to hydrophone penetrator using an aluminium U frame. All components are from the local hardware store.



Alternative mounting

To avoid the mounting of the recorder components (electronic and batteries), alternative methods can be used. This may also be necessary if the penetrator is too short to accept two nuts.

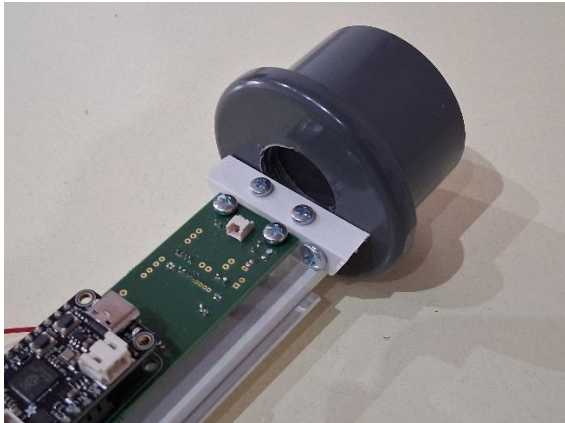


Figure 9: using PVC and ALU bars mounted on endcap (B)

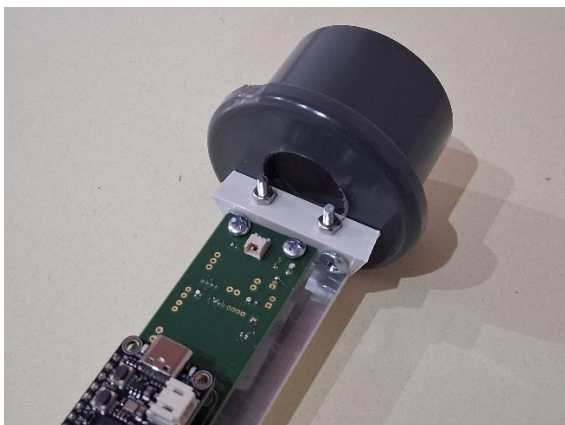


Figure 10: Using only ALU bar mounted on endcap (B)

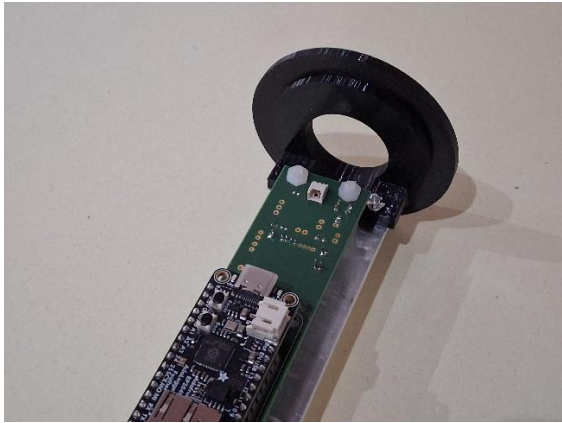


Figure 11: Using only ALU bar mounted on an 3D printed mounting structure.